

Songyu Ye

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Education

- 2025–2030 **University of California, Berkeley:** Ph.D. Mathematics
 Advisor: Professor Constantin Teleman
- 2021–2024 **Cornell University:** B.A. Summa Cum Laude Mathematics, Computer Science
 Senior Thesis: Moment Maps and Equivariant Cohomology w/ Professor Tara Holm

Interests

Representation theory, algebraic geometry, and mathematical physics, with particular interests in loop groups, affine Lie algebras, and the moduli of G -bundles on algebraic curves.

Publications

- Ye, S. (2024). *Representations of complex tori and $GL(2, \mathbb{C})$* . *Columbia Journal of Undergraduate Mathematics*, 1(1). <https://journals.library.columbia.edu/index.php/cjum/article/view/12908>
- Bumpus, M., Dai, C., Gezalyan, A., Munoz, S., Santhoshkumar, S., Ye, S., Mount, D. (2023). *Software and Analysis for Dynamic Voronoi Diagrams in the Hilbert Metric*. Canadian Conference on Computational Geometry 2023.

Talks

- Topological Field Theory Seminar (Berkeley, FA 2025): *Geometric invariant theory and derived categories*
- Toric Varieties Seminar (Cornell, SP 2024): *Line bundles on toric varieties and their cohomology*
- 4 Manifolds Seminar (Cornell, FA 2023): *Handle slides and cancellation*

Travel

- SwissMAP Research Station (SUM 2025) - Topological Symmetries and Defects in Quantum Field Theory
- IMPAN Będlewo, Poland (SUM 2025) - Torus Actions and Characteristic Classes
- Fields Institute (FA 2024) - Workshop on Toric Topology
- Park City Mathematics Institute (SUM 2024) - Summer School in Motivic Homotopy Theory
- Isaac Newton Institute for Mathematical Sciences (SUM 2024) - Moduli Stacks & Enumerative Geometry
- University of Michigan (SP 2024) - Singularities in Ann Arbor
- University of Notre Dame (SUM 2022) - Geometry and Topology Workshop: Algebraic Curves

Teaching Experience

- Graduate Student Instructor (FA 25) Math 51 Calculus 1
- Teaching Assistant (FA 23) CS 4820 Analysis of Algorithms
- Head Teaching Assistant (SP 23) CS 2802 Honors Discrete Structures
- Teaching Assistant (FA 22) CS 2800 Discrete Structures